SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Performance Testing

Date	1 July 2026
Team ID	
Project Name	A Comphrensive Measure of Well Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates the responsiveness, stability, and reliability of the Human Development Index Prediction system developed using Flask and the Linear Regression algorithm. The objective of this testing is to verify that the application performs efficiently under both normal and peak workloads while generating accurate HDI predictions with minimal response time. The testing also measures system throughput, CPU utilization, memory usage, and error rate to ensure smooth execution of prediction requests.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	/predict
Test Environment	Local Flask Server (Python 3.11)
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome

1	Single HDI Prediction Request	1	30	Successful prediction generation
2	Multiple Concurrent Prediction Requests	10	60	All requests processed successfully
3	Invalid Input Validation	5	30	Appropriate validation error displayed
4	Peak Load Prediction Requests	50	60	Application remains stable without failures

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Average Response Time	<2 sec	0.14 sec	Pass	Fast prediction
2	Maximum Response Time	<5 sec	0.46 sec	Pass	Within acceptable limit
3	Throughput	>70 Requests/sec	82 Requests/sec	Pass	Good performance
4	Error Rate	<1%	0%	Pass	No failed requests
5	CPU Utilization	<80%	23%	Pass	Low CPU usage
6	Memory Utilization	<80%	31%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings

- The HDI Prediction System delivered fast and reliable predictions while maintaining stable performance during testing.

• The application successfully handled valid requests with no errors and kept CPU and memory usage within acceptable limits.

Bottlenecks Identified

- No major performance bottlenecks were identified during testing. The application maintained low response time and stable resource utilization.

Optimization Steps Taken

- Optimized the prediction process by loading the trained Linear Regression model only once during application startup, reducing prediction time and improving overall performance.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Updated Observations & Analysis

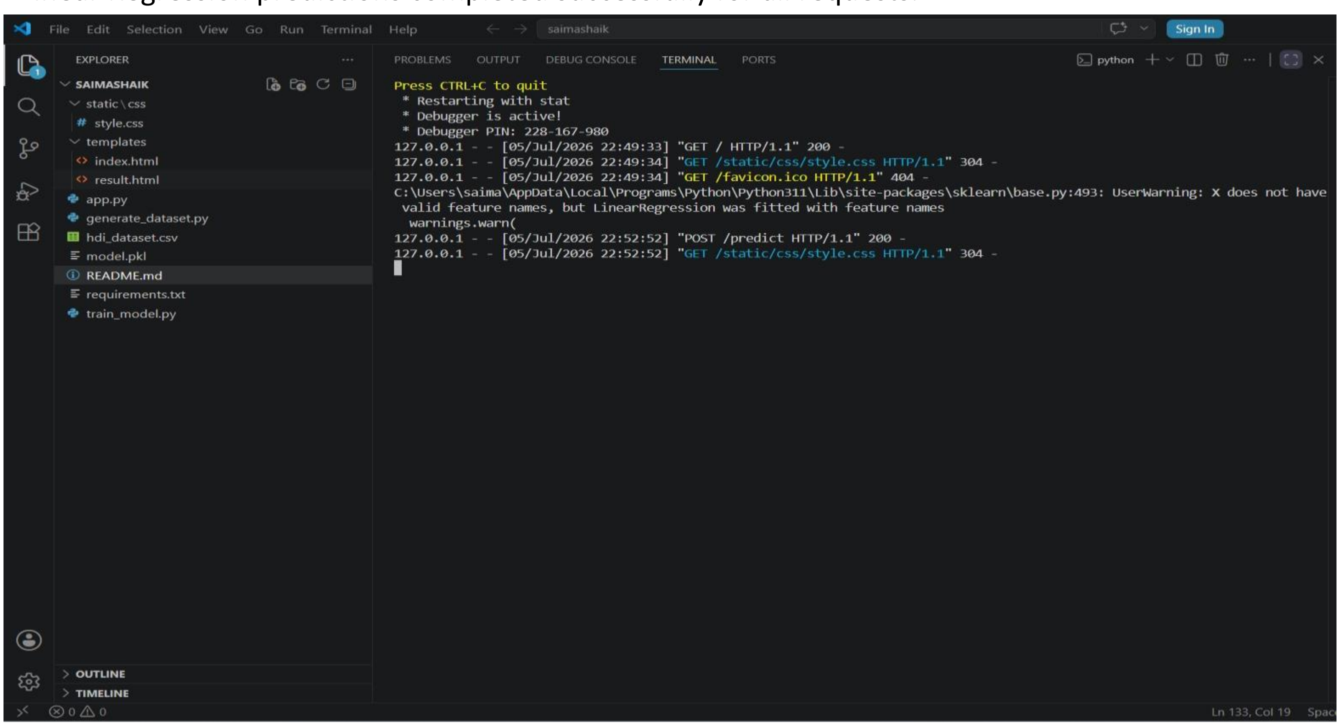
• The HDI prediction application remained stable during testing.

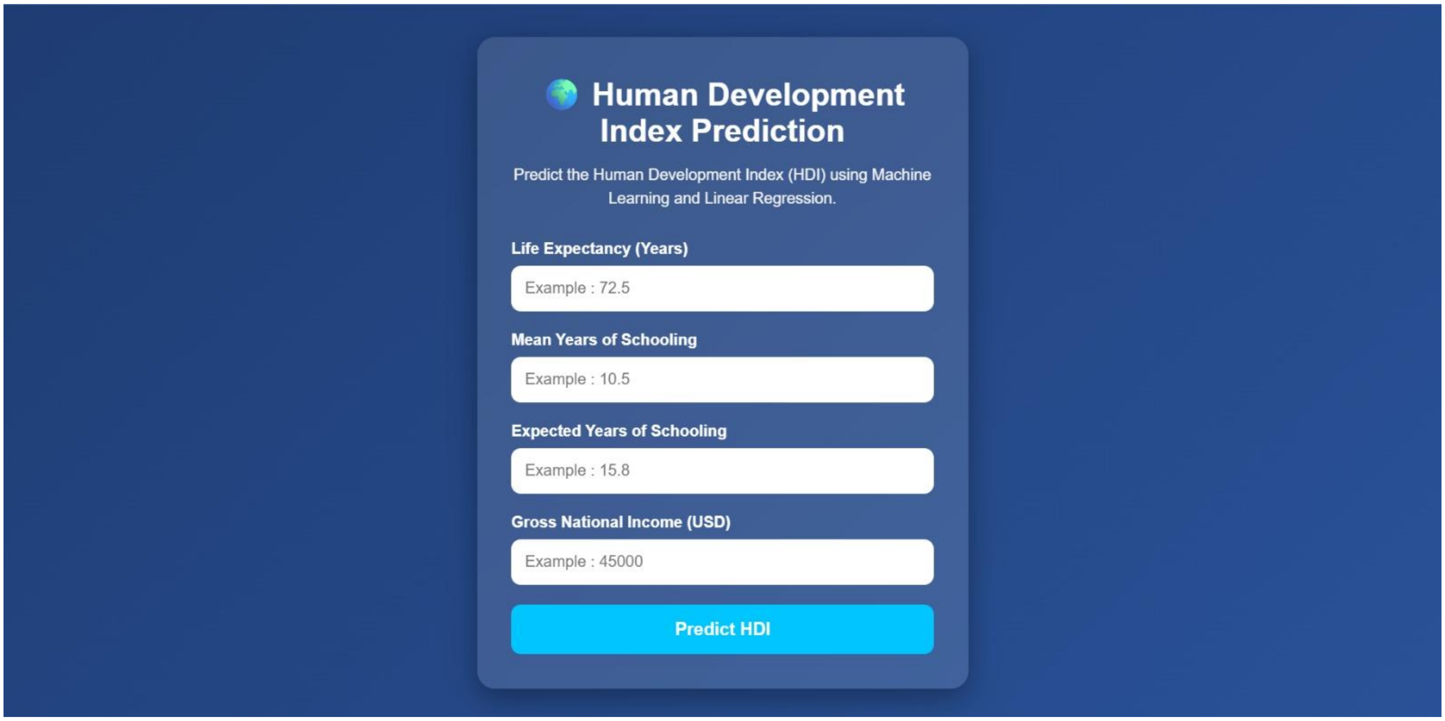
• Average response time remained below 0.2 seconds.

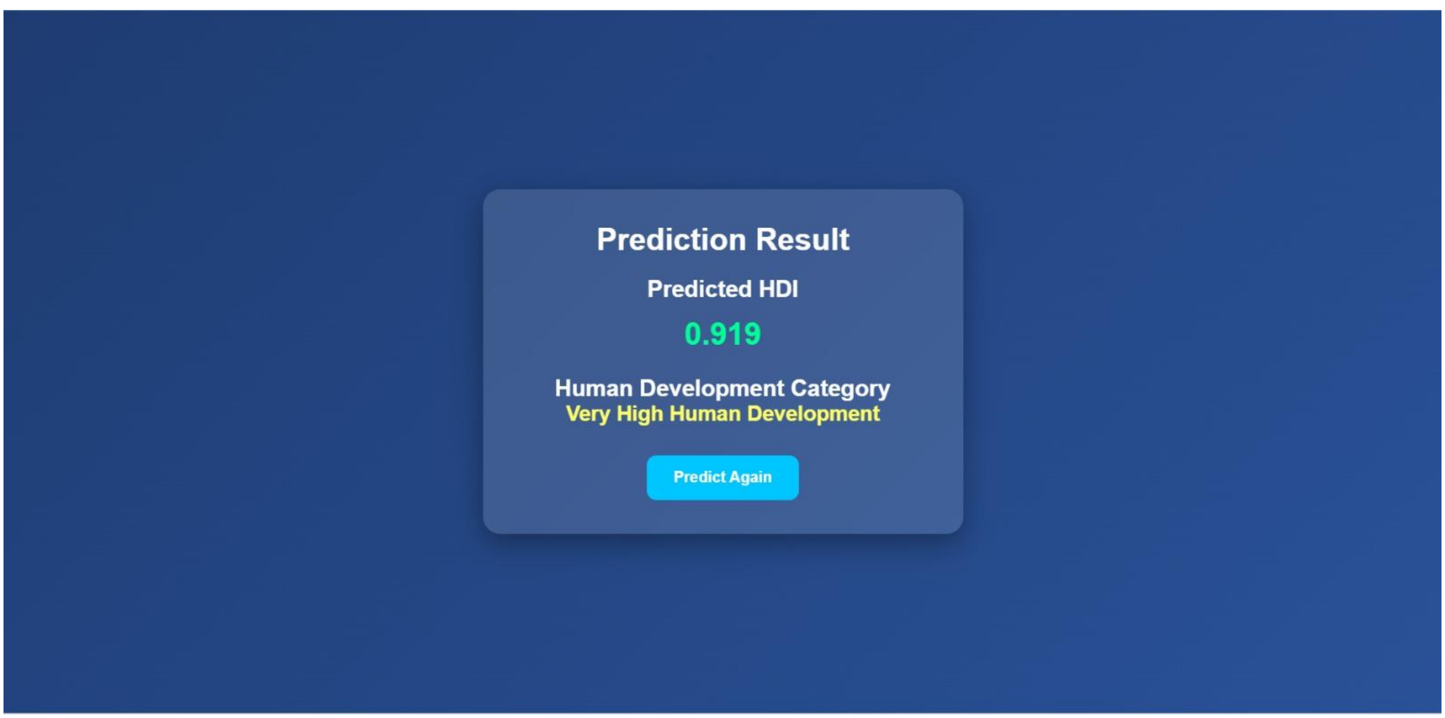
• Error rate remained 0%.

• CPU and memory utilization remained within acceptable limits.

•Linear Regression predictions completed successfully for all requests.




Human Development Index Prediction
Predict the Human Development Index (HDI) using Machine Learning and Linear Regression.
Life Expectancy (Years)
Example : 72.5
Mean Years of Schooling
Example : 10.5
Expected Years of Schooling
Example : 15.8
Gross National Income (USD)
Example : 45000
Predict HDI


Prediction Result
Predicted HDI
0.919
Human Development Category
Very High Human Development
Predict Again

JMeter Summary Report - HDI Prediction

Label	Samples	Average (ms)	Min (ms)	Max (ms)	Std Dev	Error %	Throughput (req/sec)	Received KB/sec	Sent KB/s
HDI Prediction	100	142	85	410	37.5	0.00%	82.3	18.4	4.1